

OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP - 09: CONTIGUOUS MEMORY ALLOCATION

First Fit

```
user123@localhost:~/memory
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi firstFit.c
[user123@localhost memory]$
```

```
user123@localhost:~/memory — /usr/bin/vim firstFit.c
#include <stdio.h>

int main(){
    int n, m;

    printf("Enter number of memory blocks:\n");
    scanf("%d",&m);

    int blockSize[m], remainingSize[m];

    printf("Enter size of each block:\n");
    for(int i=0; i<m; i++){
        scanf("%d", &blockSize[i]);
        remainingSize[i] = blockSize[i];
    }

    printf("Enter number of processes:\n");
    scanf("%d",&n);

    int processSize[n];

    printf("Enter size of each process:\n");
    for(int i=0; i<n; i++){
        scanf("%d",&processSize[i]);
    }

    for(int i=0; i<n; i++){
        int allocated = 0;

        for(int j=0; j<m; j++){
            if(remainingSize[j]>=processSize[i]){
                int fragment = remainingSize[j] - processSize[i];

                printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], j+1, blockSize[j],
fragment);

                remainingSize[j] -= processSize[i];

                allocated = 1;
                break;
            }
        }

        if(!allocated){
            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
        }
    }

    return 0;
}

-- INSERT --
```

```
user123@localhost:~/memory
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi firstFit.c
[user123@localhost memory]$ gcc firstFit.c
[user123@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 2 of size 500 with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with Fragment 176
Process 4 of size 426 is not allocated
[user123@localhost memory]$
```

Best Fit

```
user123@localhost:~/memory
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi bestFit.c
[user123@localhost memory]$
```

```
user123@localhost:~/memory — /usr/bin/vim bestFit.c
#include <stdio.h>

int main(){
    int m, n;

    printf("Enter number of memory blocks:\n");
    scanf("%d",&m);

    int blockSize[m], remaining[m];

    printf("Enter size of each block:\n");
    for(int i=0; i<m; i++){
        scanf("%d", &blockSize[i]);
        remaining[i] = blockSize[i];
    }

    printf("Enter number of processes:\n");
    scanf("%d",&n);

    int processSize[n];

    printf("Enter size of each process:\n");
    for(int i=0; i<n; i++){
        scanf("%d", &processSize[i]);
    }

    for(int i=0; i<n; i++){
        int bestIndex = -1;

        for(int j=0; j<m; j++){
            if(remaining[j] >= processSize[i]){
                if(bestIndex== -1 || remaining[j] < remaining[bestIndex]){
                    bestIndex = j;
                }
            }
        }

        if(bestIndex != -1){
            int fragment = remaining[bestIndex] - processSize[i];
            printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], bestIndex+1, blockSize[bestIndex], fragment);
            remaining[bestIndex] -= processSize[i];
        } else {
            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
        }
    }

    return 0;
}
-- INSERT --
```

```
user123@localhost:~/memory
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi bestFit.c
[user123@localhost memory]$ gcc bestFit.c
[user123@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 4 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with Fragment 174
[user123@localhost memory]$
```